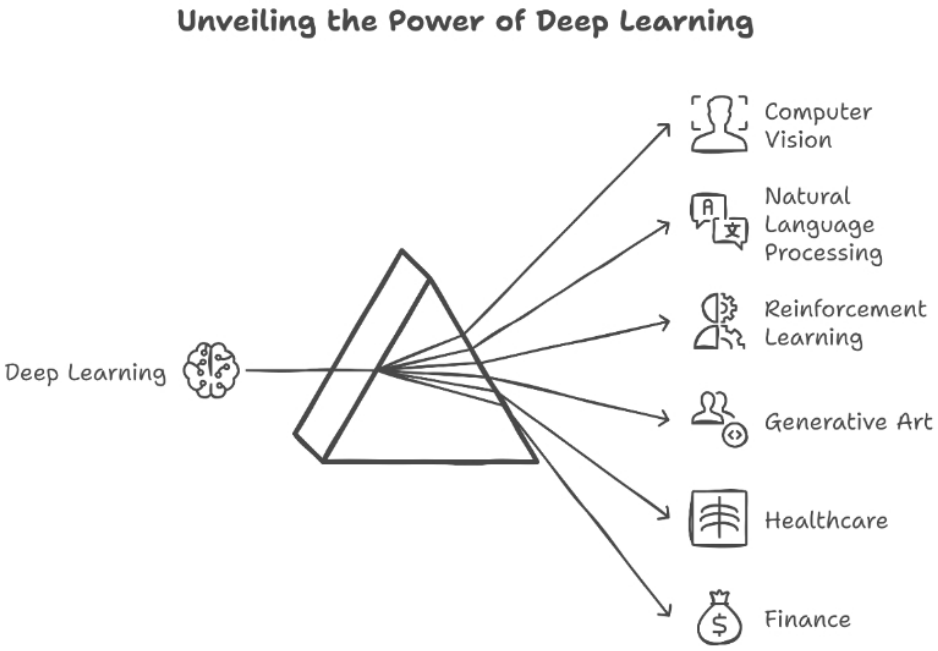
6. Deep Learning with Python

Deep learning is a transformative subset of machine learning that emphasizes algorithms modeled after the structure and function of the human brain. These sophisticated algorithms, known as neural networks, are designed to process large amounts of data and extract meaningful patterns, enabling them to perform tasks that were once considered exclusively human. Fields such as computer vision, natural language processing, reinforcement learning, and even generative art have been revolutionized by the advancements in deep learning. The ability of these networks to learn from raw data and make predictions has opened new possibilities across industries, including healthcare, finance, and autonomous systems. This chapter provides a comprehensive introduction to the foundational concepts of deep learning and practical implementations using Python, making it accessible for both beginners and experienced practitioners.



Introduction to Neural Networks

Neural networks are at the core of deep learning, a subset of machine learning that focuses on algorithms inspired by the structure and